

Addiction

Addiction stems from a chronic dysfunction of a brain system that regulates reward, motivation, and memory. A person suffering from an addiction craves a substance or behavior, often with no concern at the time about the consequences of pursuing it.

An addiction involves the repeated use of, or engagement with, a substance or activity for feelings of pleasure. Psychological and social symptoms include many behaviors, such as lack of self-control, obsession, and risk-taking. Common physical symptoms are changes in appetite, appearance changes, sleeplessness, injury or disease caused by substance abuse, and increased tolerance to the source of the addiction so that more and more of it is required

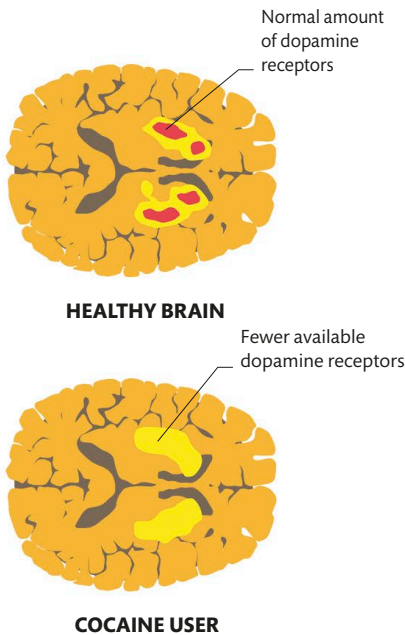
to achieve the same amount of pleasurable reward. Removal of the addiction source causes reactions such as sweating, trembling, vomiting, and behavioral changes.

Chemical pleasure

Addiction affects the brain's structure as well as how it functions. Humans feel excitement and pleasure when the brain releases neurotransmitters like dopamine, followed by a feeling of intense satisfaction from hormones such as endorphins. Endorphins relieve stress and pain in ways similar to drugs such as cocaine.

For many people, creative or physical activities, such as playing a musical instrument or exercising, release enough neurotransmitters to provide pleasure and satisfaction. For others, however, certain drugs, alcohol, and risk-taking activities such as gambling induce a quicker and much more extreme form of pleasure before eventually disrupting and damaging normal neurotransmitter circuitry.

Such artificial stimuli flood the brain with dopamine then create feelings of intense satisfaction once endorphins are released. The resulting "high" is registered by the hippocampus as a long-term memory, which leads to an urge to repeat the experience. Once this desire overrides normal behavior and the ability to function, it is classed as an addiction.



Cocaine use and dopamine

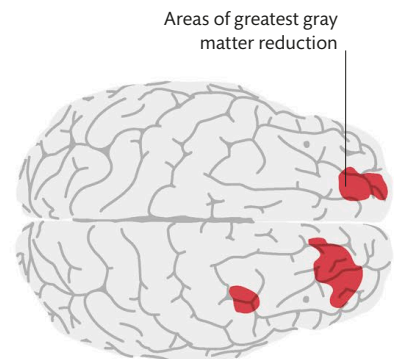
Using cocaine reduces the availability of receptors for the neurotransmitter dopamine. The result is that, over time, the user has to consume more of the drug to achieve the same sensation of reward.

TO WHAT EXTENT IS ADDICTION INHERITED?

Studies involving twins and adopted individuals show that about 40–60 percent of susceptibility to an addiction is inherited.

Why people are susceptible to addiction is not fully understood, but evidence suggests that genetic makeup may be a factor in some cases. Genes, after all, dictate not just how we respond to substances but what reactions occur when those substances are withdrawn. This may explain why some people become more readily dependent on alcohol, for example, than others.

Evaluating individuals for a suspected addiction includes the use of diagnostic tests as well as psychological assessments. They are then referred to specialists for treatment and rehabilitation.



Gray matter and methamphetamine

The use of methamphetamine shrinks the amount of gray matter in the brain's frontal cortex, among other areas, leading to a decline in mental function.